

Recommended Android PDF Export Settings

When converting wide Markdown architecture maps, box-drawing diagrams, and multi-column tables to PDF on an Android device (via browser print preview, Obsidian, Markor, or a markdown viewer):

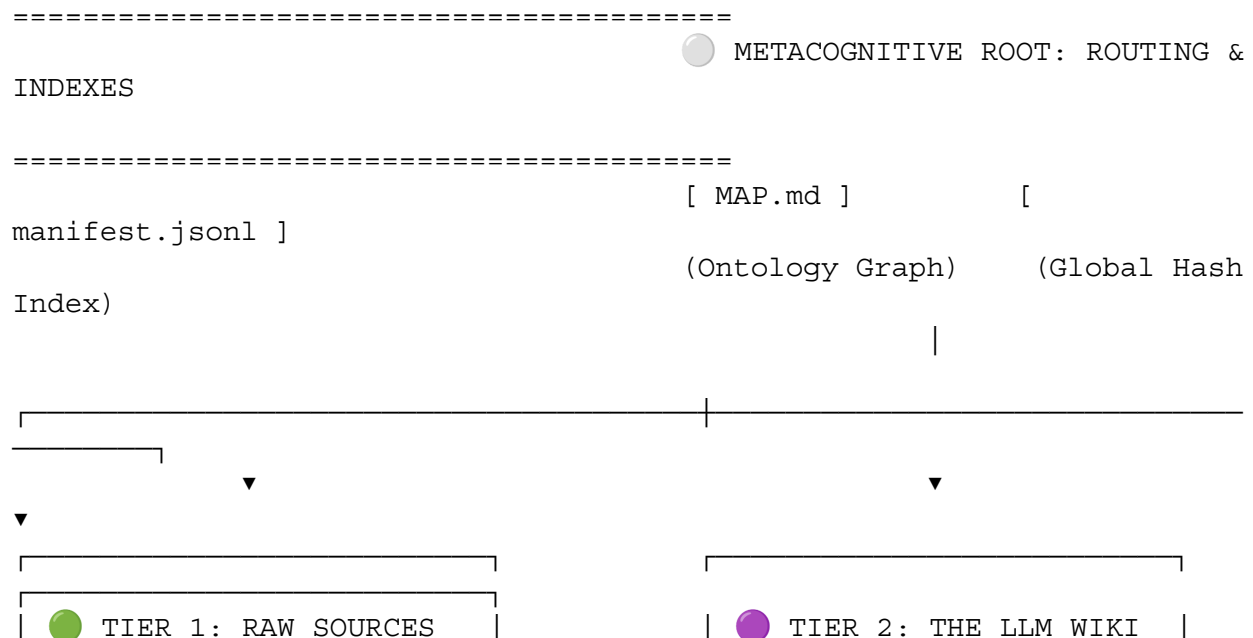
- **Page Orientation: Landscape (Mandatory).** Portrait orientation forces line wrapping on monospace code blocks wider than 80 characters, which will distort ASCII diagrams and tables.
- **Paper Size: Letter Landscape** (8.5 × 11 in / 216 × 279 mm) or **A4 Landscape** (210 × 297 mm).
 - If your PDF converter supports larger engineering paper sizes and you want the entire system schematic on a single uninterrupted poster view, select **Tabloid / Ledger (11 × 17 in)** or **A3 Landscape**.
- **Margins:** Set to **Minimum** or **None** (e.g., 0.25 in / 5 mm).
- **Scaling:** Set between **70% and 80%** (or select **"Fit to Page Width"**). This ensures that 100-character-wide code frames and complex tables render with zero line breaks.

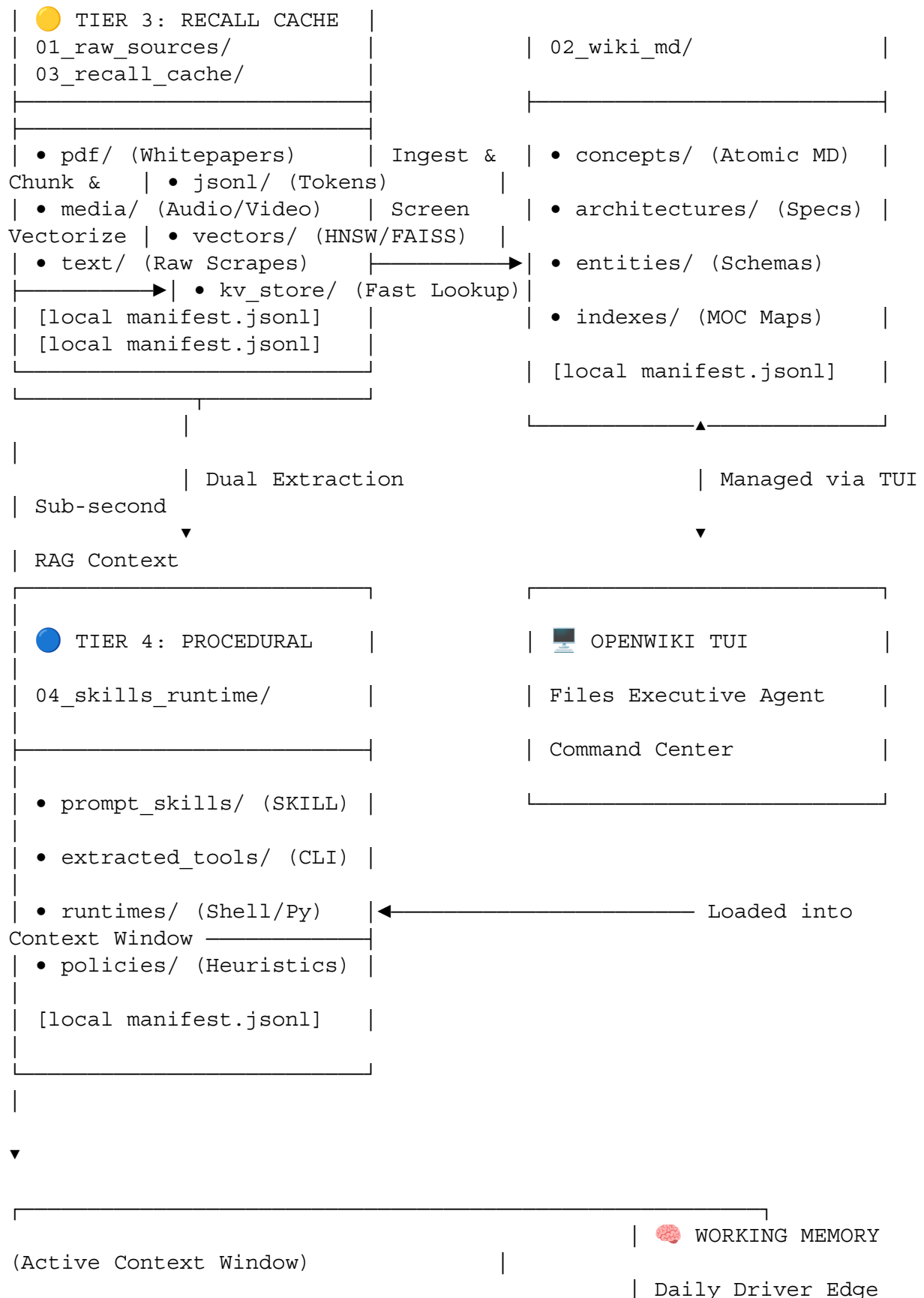


HUMAN VISUAL ARCHITECTURE MAP & VERIFICATION GUIDE

Ecosystem: 5+1 Tier Cognitive Architecture, OpenWiki TUI, and Multi-Agent RLVR **Target Audience:** System Architect / Operator

1. End-to-End System Schematic





Session

[Executor Model] [Frontier API]

[Query Model]

(Local Scripting) (Deep Reasoning)

(Task Planning)

End-of-Day P2P Sync



LOGS & HOME CROSS-AUDIT

TIER 5: EPISODIC

05_episodic_logs/

daily_driver_sync/

1. Ingest to:

2. Cross-Auditor

Agent: Reconstructs tri-model traces

3. Script Collector:

Compiles candidate tools

4. Grounding Check

against Universal Database/Memory



AUDITOR GATEKEEPER

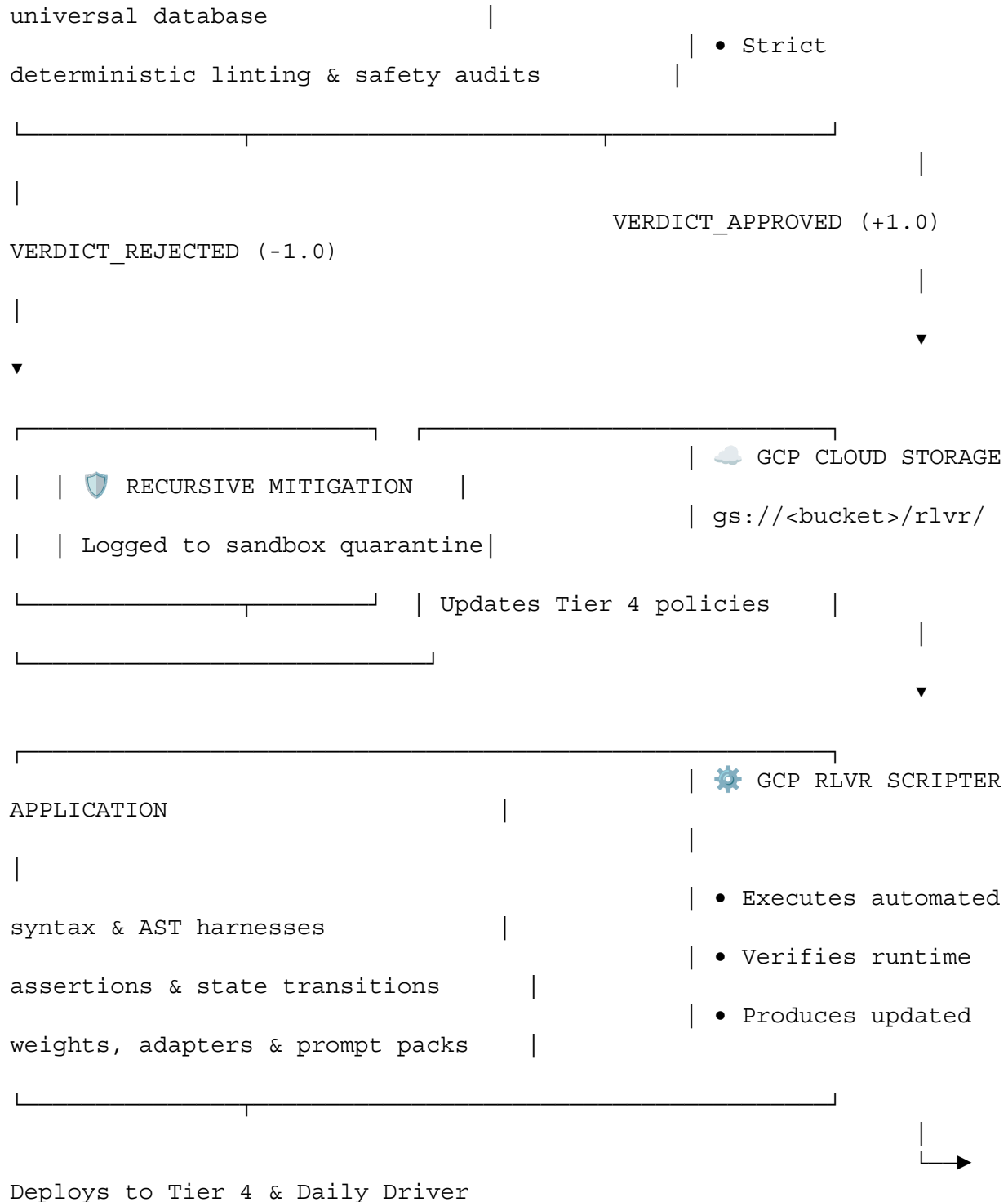
SANDBOXED RED

05_episodic_logs/red_audit_sandbox/

(No external net, clean state)

• Isolated execution

• Truth check against



2. Structural & Functional Overview

Tier	Component & Path	Operator / Engine	Primary Data Artifacts	Invariant Rule
0	Metacognitive Index /MAP.md /manifest.jsonl	Planning Agent / Global Router	Markdown link trees, JSONL file registry with SHA-256 hashes.	Must be read before performing global lookups.
1	Cold Sensory Archive 01_raw_sources/	Ingestion Engine	Raw immutable source files: .pdf, .mp4, .wav, .txt, .html.	Strictly Read-Only. Never modified by automated agents.
2	The LLM Wiki 02_wiki_md/	Files Executive Agent (<i>OpenWiki TUI</i>)	Atomic notes (concepts/), system blueprints (architectures/), registries (entities/), MOCs (indexes/).	All notes require YAML frontmatter and bidirectional links.
3	Recall Cache 03_recall_cache/	Indexing Workers	Pre-tokenized chunks (jsonl/), vector indices (vectors/), entity caches (kv_store/).	Ephemeral / Rebuildable. Discardable without data loss.
4	Procedural Runtime 04_skills_runtime/	Compilation Agent	Extracted tools (extracted_tools/), prompt templates (prompt_skills/), runtimes (runtimes/), policies (policies/).	Must enforce dual extraction (skills + executable tools).
5	Episodic Store 05_episodic_logs/	Cross-Auditor & Red Auditor Agents	Multi-model traces (trajectories/), sync batches (daily_driver_sync/), audit logs (red_audit_sandbox/), verifier scores (rlvr_verifiers/).	Append-Only. Stores telemetry and verifier reward values.

3. Human Operator Verification Guide

Use this checklist to audit the repository structure, verify synchronization pipelines, and ensure memory integrity across the nodes.

Check 1: Directory & Manifest Integrity

```
# Verify that all 6 tiers contain an active manifest.jsonl
for dir in . 01_raw_sources 02_wiki_md 03_recall_cache
04_skills_runtime 05_episodic_logs; do
```

```

if [ -f "$dir/manifest.jsonl" ]; then
    echo "PASS: $dir/manifest.jsonl exists ($(wc -l <
"$dir/manifest.jsonl") records)"
else
    echo "FAIL: $dir/manifest.jsonl missing!"
fi
done

```

- **Pass Criteria:** All 6 manifest files exist and are populated with valid JSON objects containing id, path, tier, sha256, and last_synced.

Check 2: The LLM Wiki (02_wiki_md/) Link Graph

1. Open the repository inside the **OpenWiki TUI**.
2. Run a graph health check to ensure there are no broken links:
 - Every link formatted as `[[concept_name]]` must resolve to an existing .md file under 02_wiki_md/.
 - Check frontmatter integrity: Confirm that `sources:`, `extracted_skills:`, and `extracted_tools:` arrays contain valid file paths.
3. Verify that 02_wiki_md/indexes/ contains up-to-date Maps of Content (MOCs) reflecting newly ingested concepts.

Check 3: Dual Skill & Tool Extraction Coverage

When a new technical source document (e.g., manual.pdf) is added:

- **Tool Extraction Check:** Inspect 04_skills_runtime/extracted_tools/cli/ or wrappers/ to confirm that any command syntax or code snippets from the PDF were extracted into executable .sh or .py files.
- **Skill Extraction Check:** Inspect 04_skills_runtime/prompt_skills/ to confirm that operational workflows or reasoning patterns were formatted into a SKILL.md with standard YAML frontmatter.
- **Traceability Check:** Verify that the provenance_source field in 04_skills_runtime/manifest.jsonl correctly references the original document in 01_raw_sources/.

Check 4: End-of-Day P2P Sync & Cross-Audit Trace Alignment

1. **Sync Check:** Confirm that daily driver session logs are transferred via P2P into 05_episodic_logs/daily_driver_sync/YYYYMMDD_session.jsonl.
2. **Trace Alignment Check:** Verify that each session record interleaves:
 - **Query Model:** Intent, system context, decomposition steps.
 - **Executor Model:** Tool parameters, command executions, return codes.
 - **Frontier Model:** Fallbacks, complex refactors, escalation reasoning.
3. **Database Grounding Check:** Confirm that the Cross-Auditor evaluated script outputs against the universal database before passing files to the Red Auditor.

Check 5: Sandboxed Red Auditor Gatekeeper

Inspect 05_episodic_logs/red_audit_sandbox/:

- Confirm that no script was uploaded to the cloud without a corresponding VERDICT_APPROVED record.
- Check rejected scripts in 05_episodic_logs/red_audit_sandbox/rejected_*.jsonl:
 - Each rejection must include: error_type, offending_line, sandbox_trace, and violated_invariant.
 - Verify that rejected patterns caused a corresponding update in 04_skills_runtime/policies/ to prevent recurrence.

Check 6: Cloud RLVR Pipeline & Storage Sync

- **Storage Ingestion:** Check Google Cloud Storage (gs://<repo>-rlvr-training-pipeline/) for incoming verified batches.
- **Reward Distribution:** Inspect 05_episodic_logs/rlvr_verifiers/:
 - Tasks that pass deterministic unit tests, syntax checks, and state validations must register reward: 1.0.
 - Failed attempts must register reward: -1.0 alongside AST error traces.
- **Model Deployment:** Verify that new model adapters, weights, or refined prompt skill bundles generated by the cloud scripser synchronize back into 04_skills_runtime/prompt_skills/ on local nodes.

